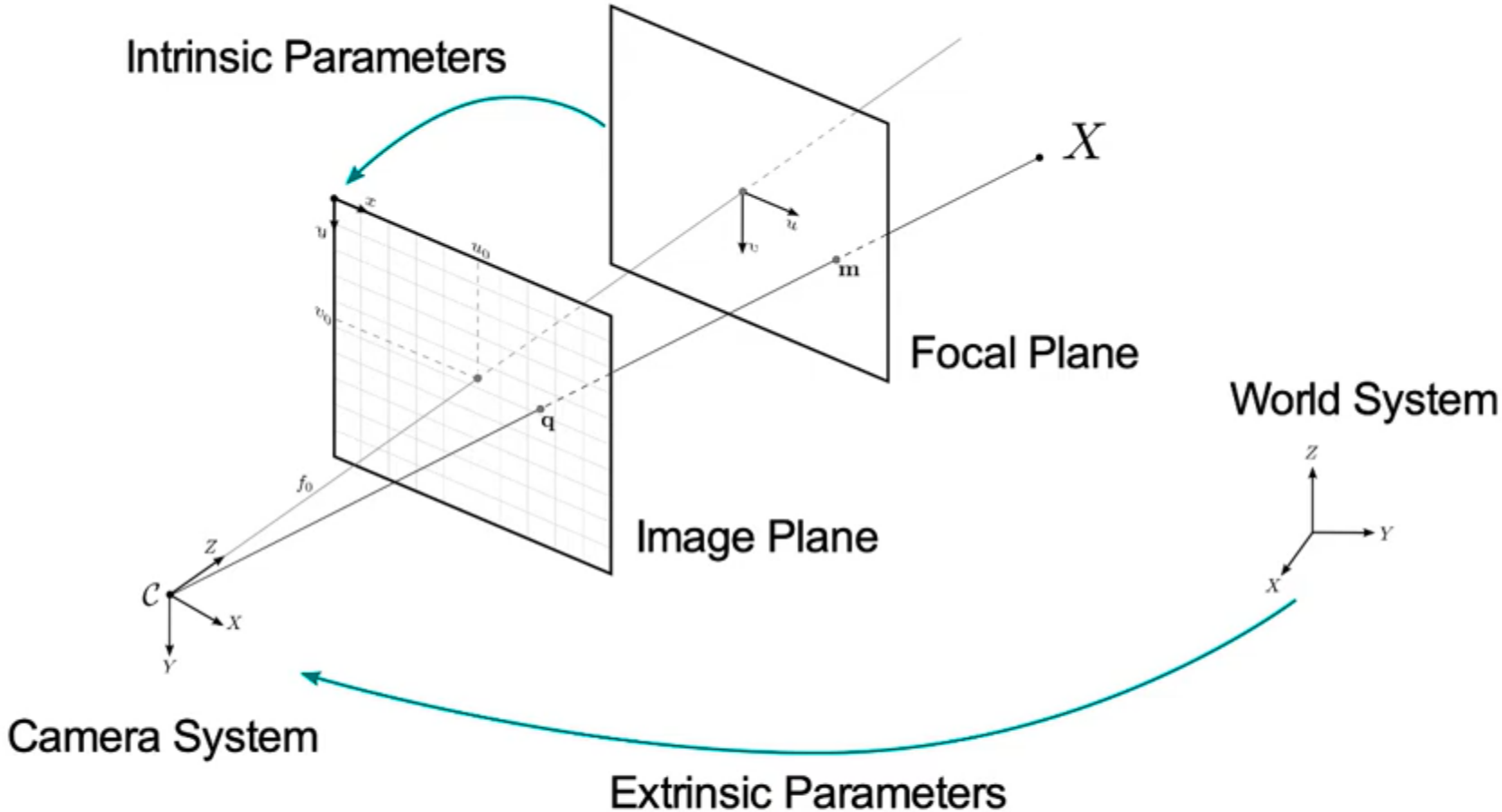
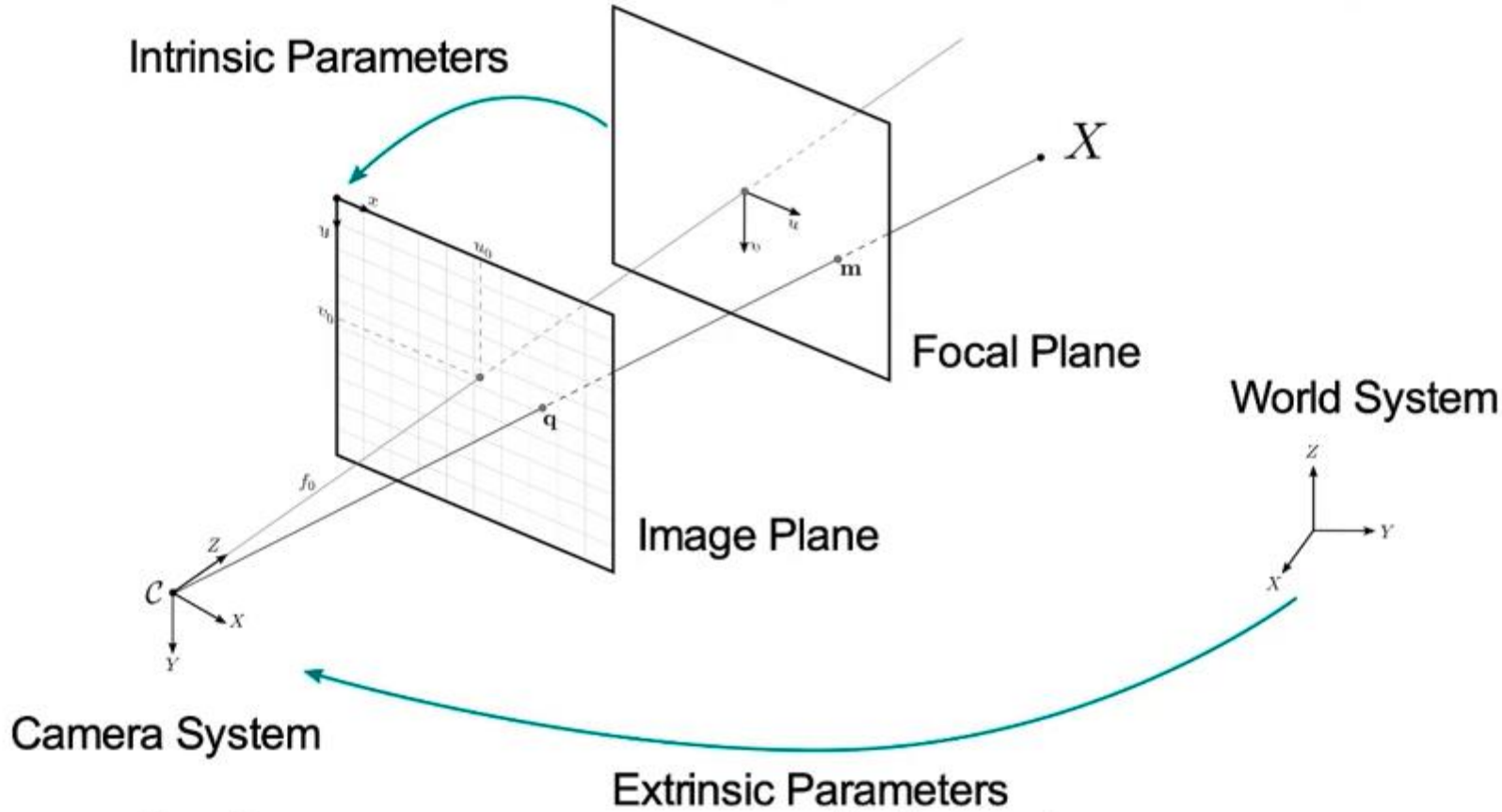
World frame to Camera frame to Pixel coordinates

Parameters



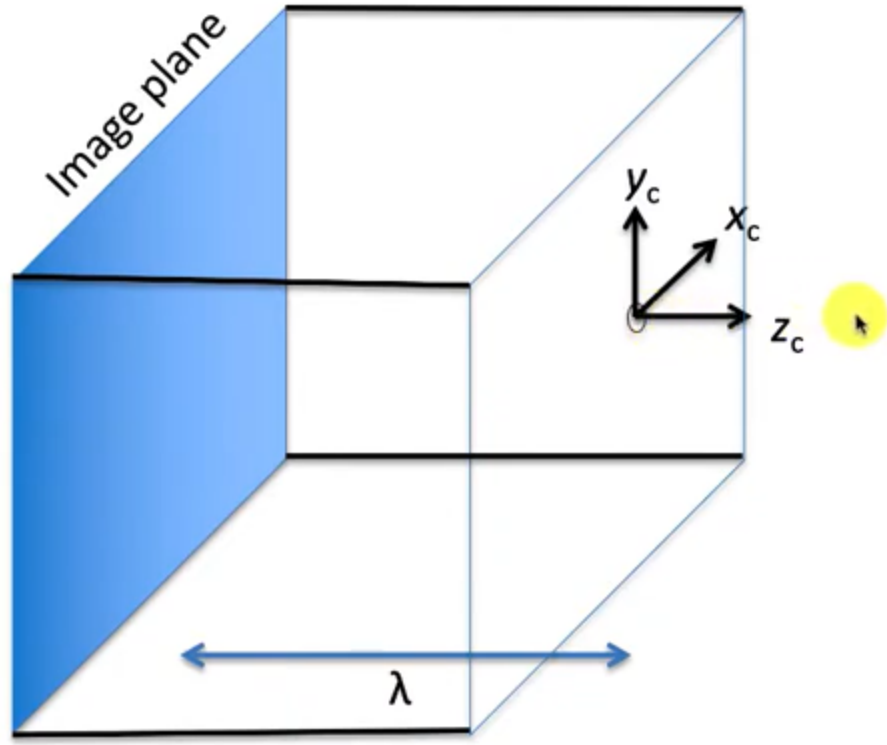
pinhole camera geometry models projective camera with two sub-parametrizations, **intrinsic** and **extrinsic** parameters

Image from http://openmvg.readthedocs.org/en/latest/openMVG/cameras/cameras/

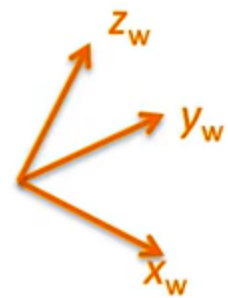


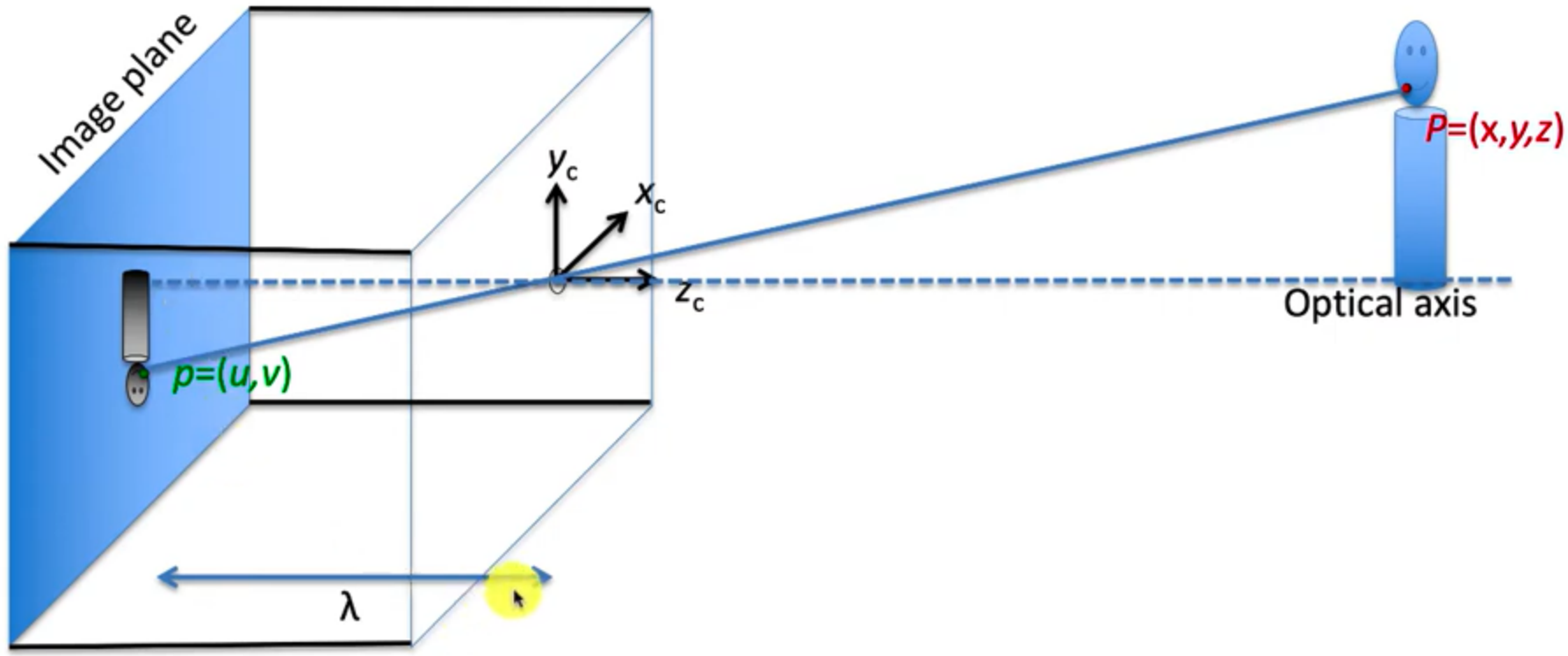
- **Intrinsic parameters** model optic component (without distortion)
- **extrinsic** model camera position and orientation in space. Extrinsic = homogenous transformation

Image from http://openmvg.readthedocs.org/en/latest/openMVG/cameras/cameras/

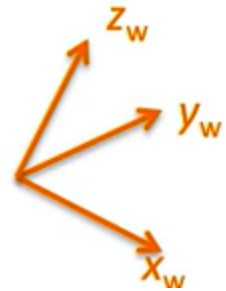


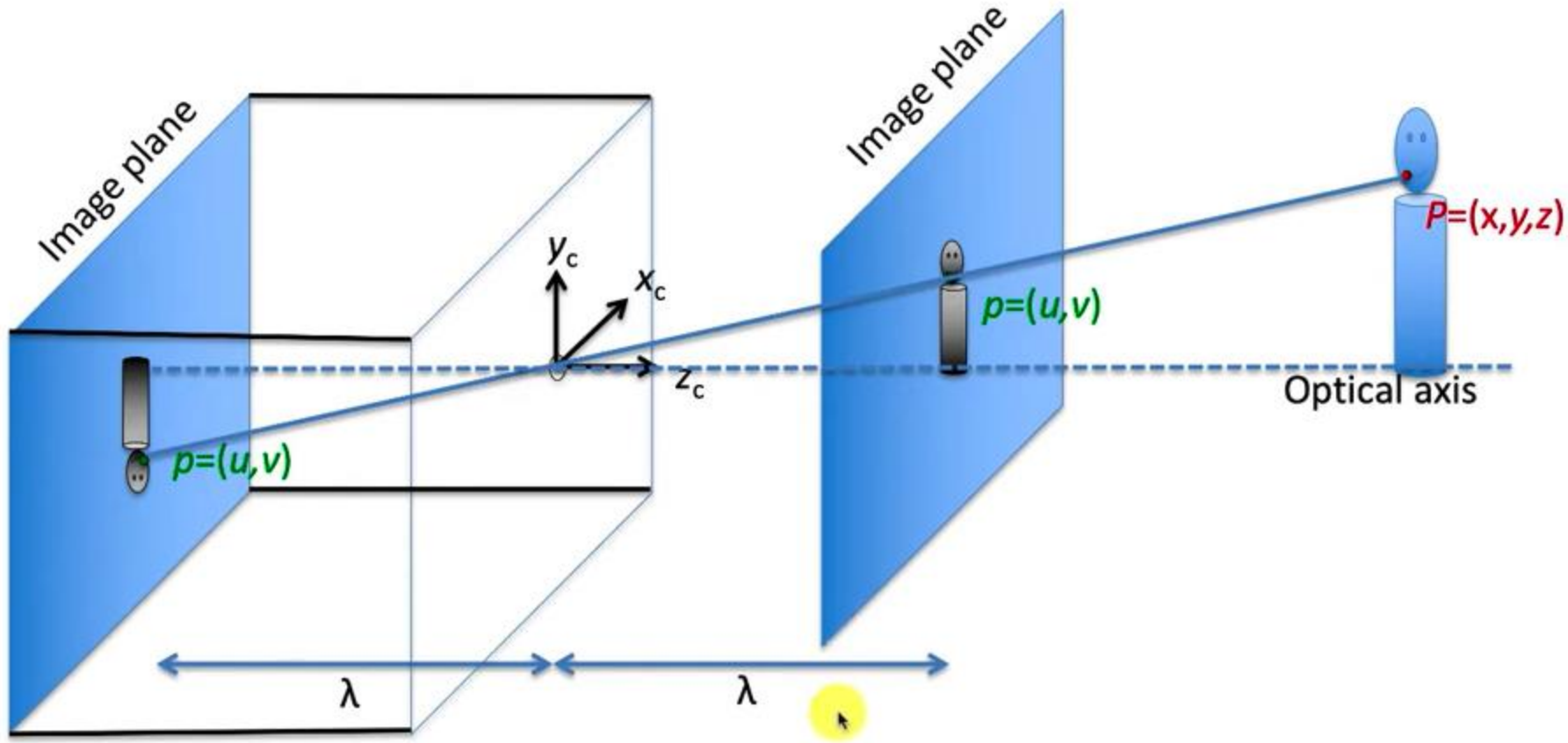
Camera (pinhole)



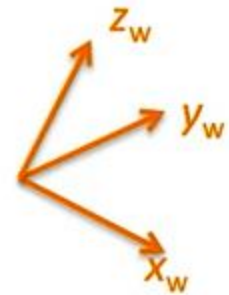


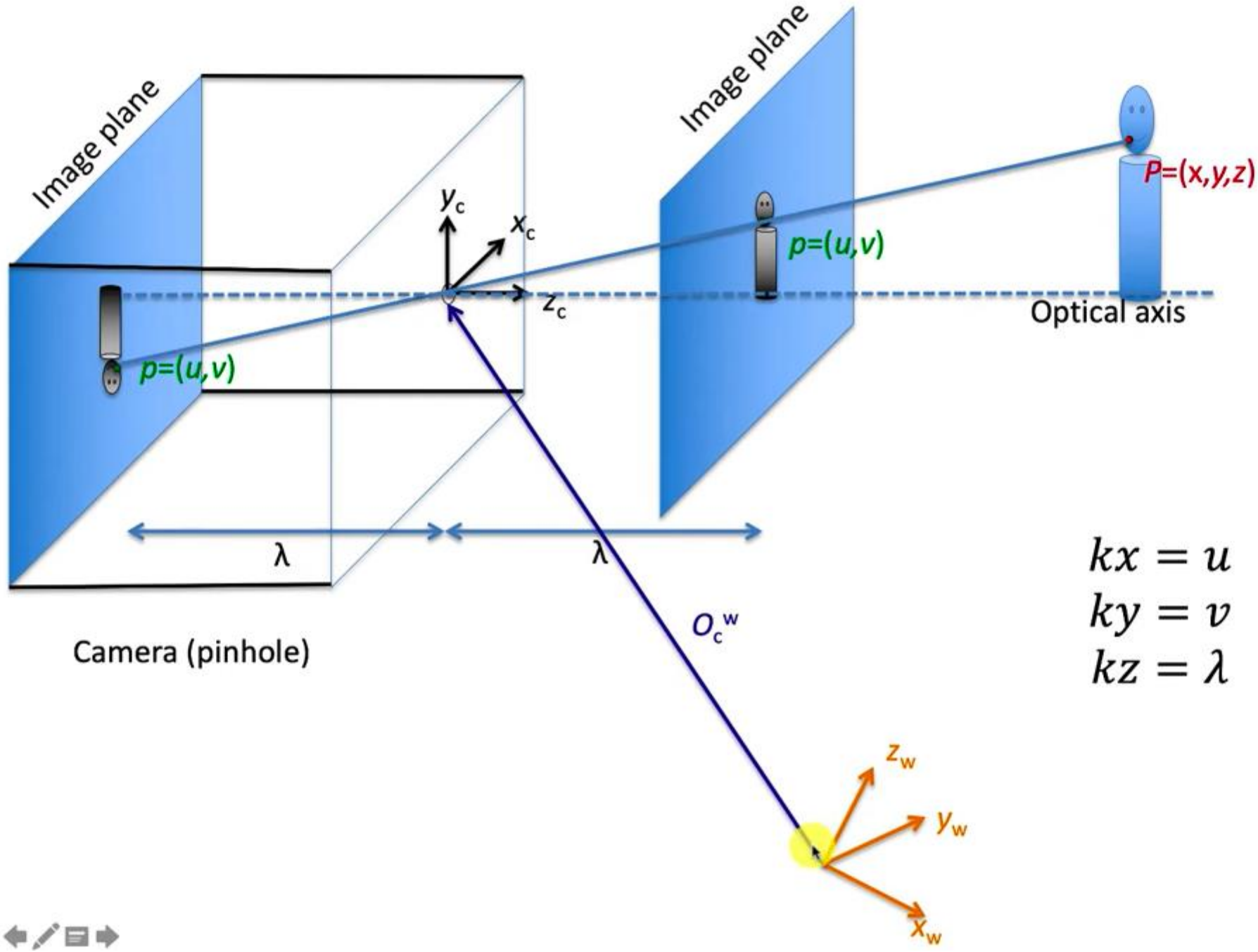
Camera (pinhole)



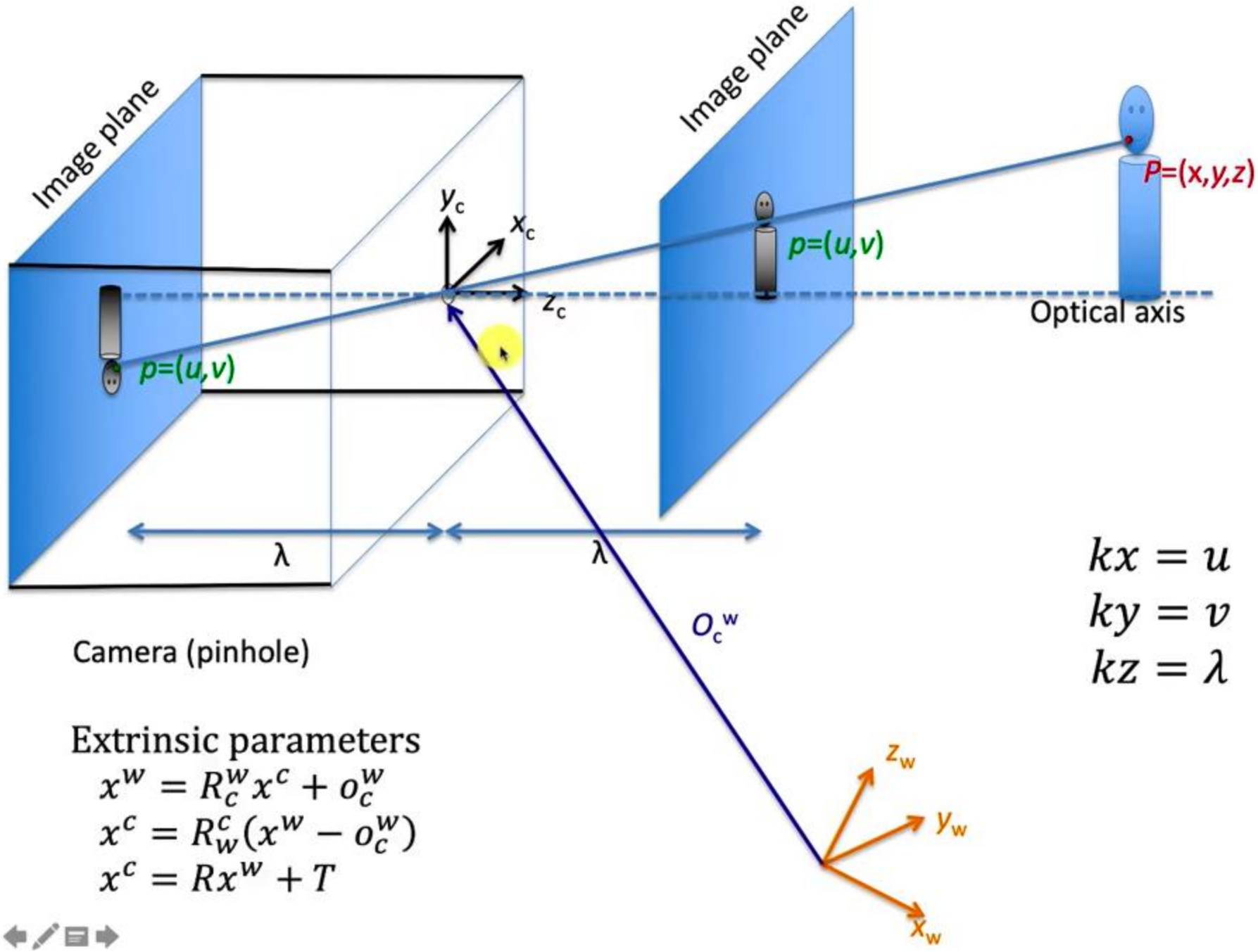


Camera (pinhole)





$$kx = u$$
$$ky = v$$
$$kz = \lambda$$



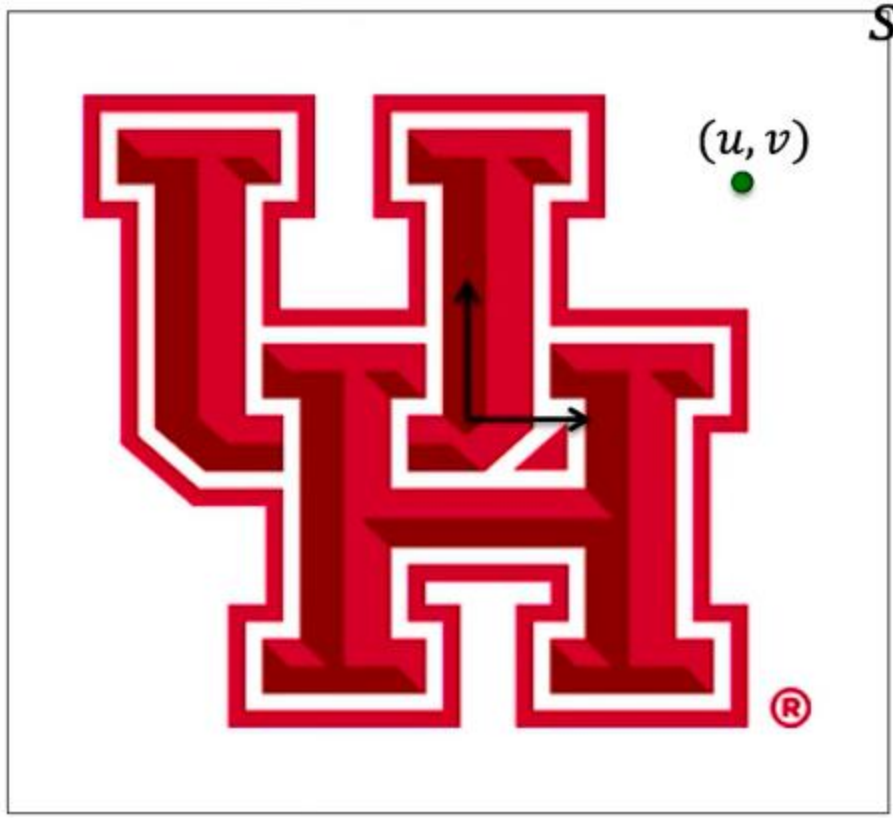
Camera (pinhole)

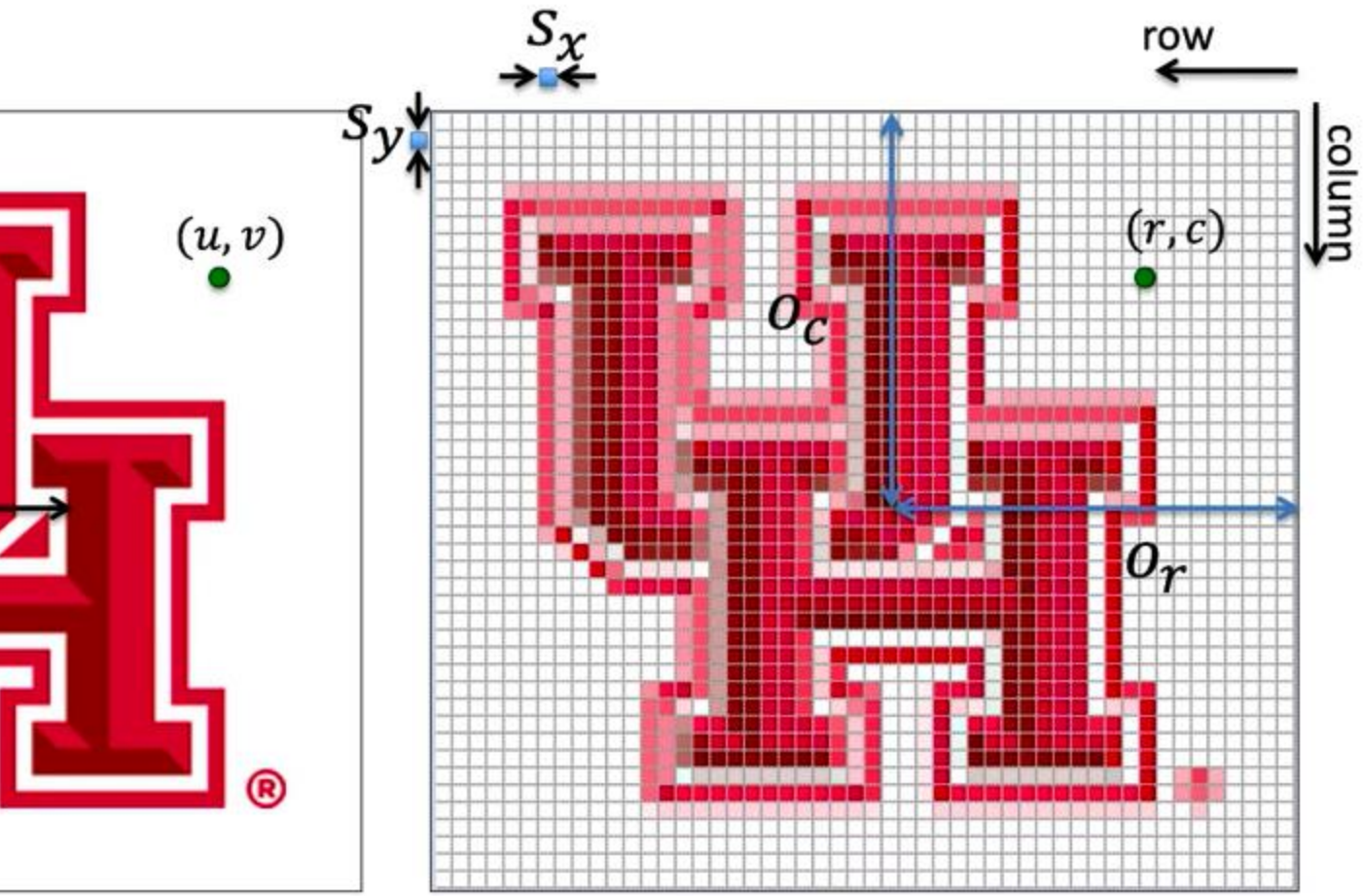
$$kx = u$$
$$ky = v$$
$$kz = \lambda$$

Extrinsic parameters

$$x^w = R_c^w x^c + o_c^w$$
$$x^c = R_w^c (x^w - o_c^w)$$
$$x^c = R x^w + T$$

Pixel coordinates (r,c) principle point at (o_r , o_c)





$$\text{Round}\left(-\frac{u}{s_x}\right) = (r - o_r), \quad \text{Round}\left(-\frac{v}{s_y}\right) = (c - o_c)$$

Camera plane→ Image frame

For a camera with focal length $\lambda = 10$, find the image plane coordinates for the 3D points whose coordinates in the camera frame are given below. Circle points not visible to a physical camera.

Transformation: $k \begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} \quad \end{bmatrix}, k =$

a. $(25,25,50)^c$ → $k =$, $[u,v] = [\quad]$

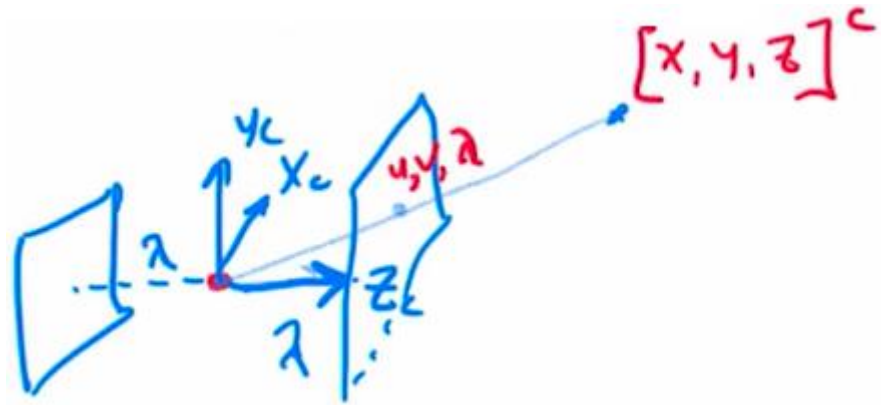
b. $(-25,-25,50)^c$ → $k =$, $[u,v] = [\quad]$

c. $(20,5,-50)^c$ → $k =$, $[u,v] = [\quad]$

d. $(15,10,25)^c$ → $k =$, $[u,v] = [\quad]$

e. $(0,0,50)^c$ → $k =$, $[u,v] = [\quad]$

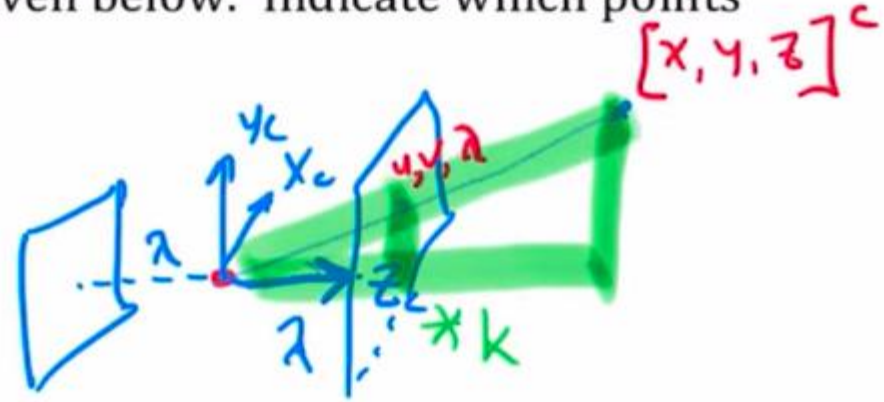
f. $(0,0,100)^c$ → $k =$, $[u,v] = [\quad]$


[x, y, z]^c
y_c
x_c
u, v, 1
λ
λ
z_c

1. For a camera with focal length $\lambda = 10$, find the image plane coordinates for the 3D points whose coordinates in the camera frame are given below. Indicate which points will not be visible to a physical camera.

Transformation: $k \begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} u \\ v \\ \lambda \end{bmatrix}$, $k = \frac{\lambda}{z^c}$

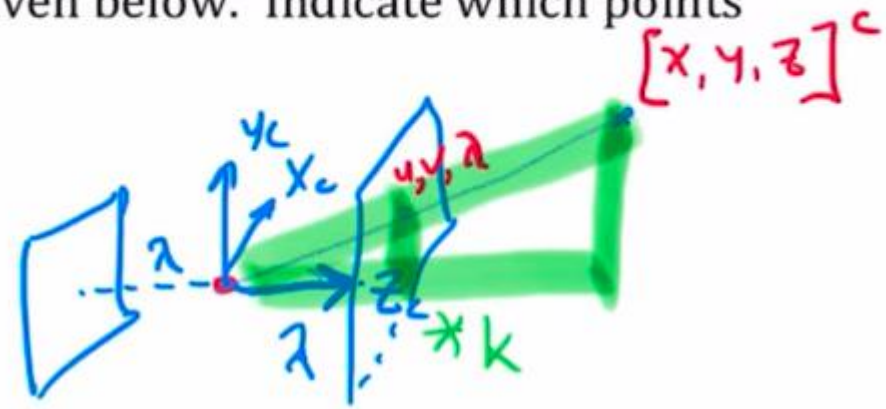
$kz^c = \lambda$ $\quad k =$



1. For a camera with focal length $\lambda = 10$, find the image plane coordinates for the 3D points whose coordinates in the camera frame are given below. Indicate which points will not be visible to a physical camera.

Transformation: $k \begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} u \\ v \\ \lambda \end{bmatrix}$, $k = \frac{\lambda}{z^c}$

$k z^c = \lambda$ $\quad k =$



a. $(25,25,50)^c \rightarrow k = \frac{1}{5}$ $\quad (u,v) = (5, 5)$ $\quad \frac{10}{50}$

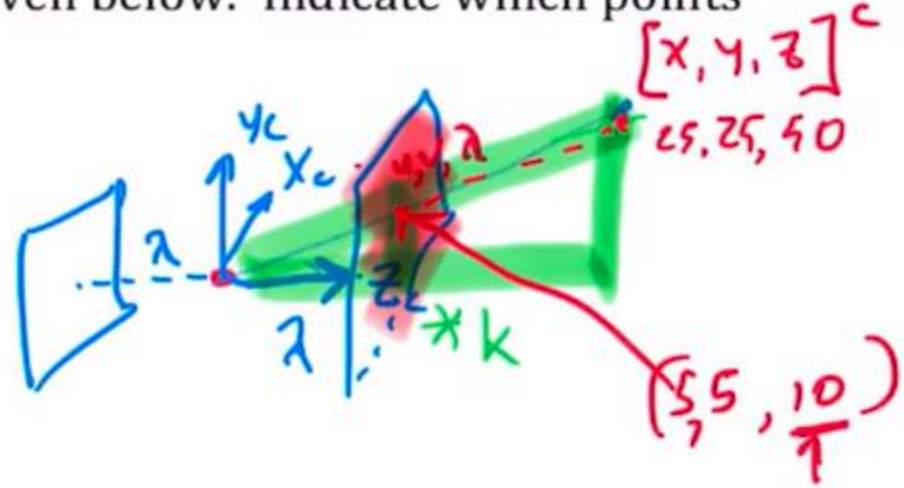
b. $(-25,-25,50)^c \rightarrow k =$ $\quad (u,v) =$

c. $(20,5,-50)^c \rightarrow k =$ $\quad (u,v) =$

1. For a camera with focal length $\lambda = 10$, find the image plane coordinates for the 3D points whose coordinates in the camera frame are given below. Indicate which points will not be visible to a physical camera.

Transformation: $k \begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} u \\ v \\ \lambda \end{bmatrix}$, $k = \frac{\lambda}{z^c}$

$k z^c = \lambda$ $\quad k =$



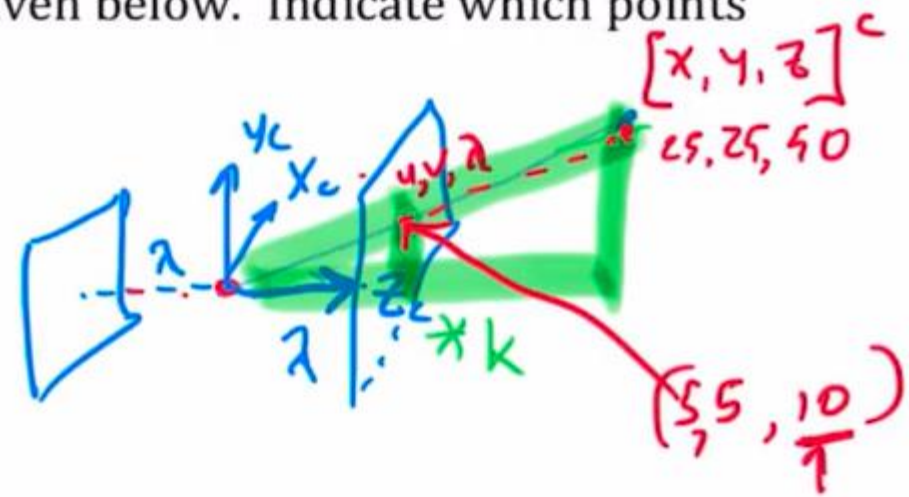
a. $(25,25,50)^c$ $\rightarrow k = \frac{1}{5}$ $,(u,v) = (5, 5)$

$\frac{10}{50}$

1. For a camera with focal length $\lambda = 10$, find the image plane coordinates for the 3D points whose coordinates in the camera frame are given below. Indicate which points will not be visible to a physical camera.

Transformation: $k \begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} u \\ v \\ \lambda \end{bmatrix}$, $k = \frac{\lambda}{z^c}$

$k z^c = \lambda$ $\quad k =$



a. $(25,25,50)^c \rightarrow k = \frac{1}{5}$, $(u,v) = (5, 5)$ $\quad \frac{10}{50}$

b. $(-25,-25,50)^c \rightarrow k = \frac{1}{5}$, $(u,v) = (-5, -5)$ $\quad \frac{10}{50}$

c. $(20,5,-50)^c \rightarrow k = \frac{-1}{5}$, $(u,v) = (-4, -1)$ $\quad = -\frac{10}{50}$

Not visible

d. $(15,10,25)^c \rightarrow k =$, $(u,v) =$

e. $(0,0,50)^c \rightarrow k =$, $(u,v) =$

f. $(0,0,100)^c \rightarrow k =$, $(u,v) =$

a. $(25,25,50)^c \rightarrow k = \frac{1}{5}$, $(u,v) = (5, 5)$ $\frac{10}{50}$

b. $(-25,-25,50)^c \rightarrow k = \frac{1}{5}$, $(u,v) = (-5, -5)$ $\frac{10}{50}$

c. $(20,5,-50)^c \rightarrow k = -\frac{1}{5}$, $(u,v) = (-4, -1)$ $= -\frac{10}{50}$

Not visible

d. $(15,10,25)^c \rightarrow k = \frac{2}{5}$, $(u,v) = (6, 4)$ $\frac{10}{25}$

e. $(0,0,50)^c \rightarrow k = \frac{1}{5}$, $(u,v) = (0, 0)$ $\frac{10}{50}$

f. $(0,0,100)^c \rightarrow k = \frac{1}{10}$, $(u,v) = (0, 0)$ $\frac{10}{100}$

2. Repeat for the case where coordinates of points given with respect to the world frame. Suppose the optical axis is parallel with the world *x*-axis, the camera *x*-axis is parallel to the world y-axis, and the center of projection has coordinates (0,0,100).

$$\begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} =$$

a. $(25,25,50)^w$ → ( )^c → $k =$, $(u,v) =$

b. $(-25,-25,50)^w$ → ( )^c → $k =$, $(u,v) =$

c. $(20,5,-50)^w$ → ( )^c → $k =$, $(u,v) =$

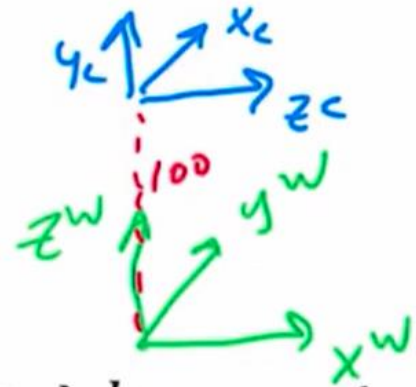
d. $(15,10,25)^w$ → ( )^c → $k =$, $(u,v) =$

e. $(0,0,50)^w$ → ( )^c → $k =$, $(u,v) =$

f. $(0,0,100)^w$ → ( )^c → $k =$, $(u,v) =$

2. Repeat for the case where coordinates of points given with respect to the world frame. Suppose the optical axis is parallel with the world *x*-axis, the camera *x*-axis is parallel to the world y-axis, and the center of projection has coordinates (0,0,100).

$$\begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} y^w \\ z^w - 100 \\ x^w \end{bmatrix}$$



a. $(25,25,50)^w$ → ( )^c → k = , (u,v) =

b. $(-25,-25,50)^w$ → ( )^c → k = , (u,v) =

c. $(20,5,-50)^w$ → ( )^c → k = , (u,v) =

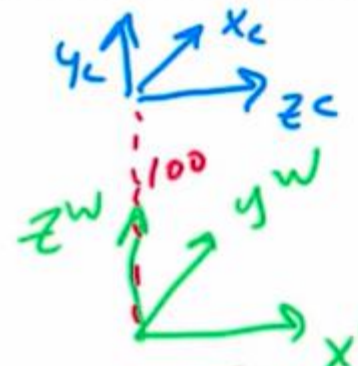
d. $(15,10,25)^w$ → ( )^c → k = , (u,v) =

e. $(0,0,50)^w$ → ( )^c → k = , (u,v) =

f. $(0,0,100)^w$ → ( )^c → k = , (u,v) =

2. Repeat for the case where coordinates of points given with respect to the world frame. Suppose the optical axis is parallel with the world x-axis, the camera x-axis is parallel to the world y-axis, and the center of projection has coordinates (0,0,100).

$$\begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} y^w \\ z^w - 100 \\ x^w \end{bmatrix}$$



a. $(25,25,50)^w \rightarrow (25,-50,25)^c \rightarrow k = \frac{2}{5}$, $(u,v) = (10,-70)$ ($\frac{10}{25}$)

b. $(-25,-25,50)^w \rightarrow (\quad)^c \rightarrow k =$, $(u,v) =$

c. $(20,5,-50)^w \rightarrow (\quad)^c \rightarrow k =$, $(u,v) =$

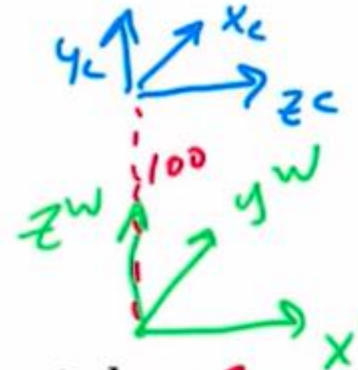
d. $(15,10,25)^w \rightarrow (\quad)^c \rightarrow k =$, $(u,v) =$

e. $(0,0,50)^w \rightarrow (\quad)^c \rightarrow k =$, $(u,v) =$

f. $(0,0,100)^w \rightarrow (\quad)^c \rightarrow k =$, $(u,v) =$

2. Repeat for the case where coordinates of points given with respect to the world frame. Suppose the optical axis is parallel with the world *x*-axis, the camera *x*-axis is parallel to the world y-axis, and the center of projection has coordinates (0,0,100).

$$\begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} y^w \\ z^w - 100 \\ x^w \end{bmatrix}$$



a. $(25,25,50)^w \rightarrow (25,-50,25)^c \rightarrow k = \frac{2}{5}, (u,v) = (10,-70)$

b. $(-25,-25,50)^w \rightarrow (-25,-50,-25)^c \rightarrow k = \frac{-2}{5}, (u,v) = (-10,-20)$ ⇒ INVISIBLE

c. $(20,5,-50)^w \rightarrow (\quad)^c \rightarrow k = \quad, (u,v) =$

d. $(15,10,25)^w \rightarrow (\quad)^c \rightarrow k = \quad, (u,v) =$

e. $(0,0,50)^w \rightarrow (\quad)^c \rightarrow k = \quad, (u,v) =$

f. $(0,0,100)^w \rightarrow (\quad)^c \rightarrow k = \quad, (u,v) =$

2. Repeat for the case where coordinates of points given with respect to the world frame. Suppose the optical axis is parallel with the world *x*-axis, the camera *x*-axis is parallel to the world y-axis, and the center of projection has coordinates (0,0,100).

$$\begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} y^w \\ z^w - 100 \\ x^w \end{bmatrix}$$



a. $(25,25,50)^w \rightarrow (25, -50, 25)^c \rightarrow k = \frac{2}{5}$ (10/25), $(u,v) = (10, -20)$

b. $(-25,-25,50)^w \rightarrow (-25, -50, -25)^c \rightarrow k = \frac{-2}{5}$ (−2/5), $(u,v) = (-10, -20)$ ⇒ INVISIBLE

c. $(20,5,-50)^w \rightarrow (5, -150, 20)^c \rightarrow k = \frac{1}{2}$ (10/20), $(u,v) = (2.5, -75)$

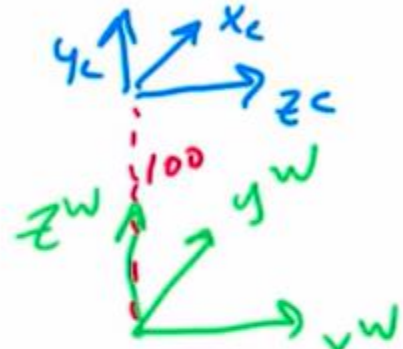
d. $(15,10,25)^w \rightarrow (10, -75, 15)^c \rightarrow k = \frac{2}{3}$ (10/15), $(u,v) = (\frac{20}{3}, -50)$

e. $(0,0,50)^w \rightarrow (0, -50, 0)^c \rightarrow k = \frac{10}{0}$, $(u,v) = (NaN, NaN)$ — INVISIBLE ONES

f. $(0,0,100)^w \rightarrow (\quad)^c \rightarrow k = \quad , (u,v) =$

2. Repeat for the case where coordinates of points given with respect to the world frame. Suppose the optical axis is parallel with the world x-axis, the camera x-axis is parallel to the world y-axis, and the center of projection has coordinates (0,0,100).

$$\begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \begin{bmatrix} y^w \\ z^w - 100 \\ x^w \end{bmatrix}$$



a. $(25,25,50)^w \rightarrow (25, -50, 25)^c \rightarrow k = \frac{10}{25} = \frac{2}{5}$, $(u,v) = (10, -20)$

b. $(-25,-25,50)^w \rightarrow (-25, -50, -25)^c \rightarrow k = \frac{-2}{5}$, $(u,v) = (-10, -20)$ ⇒ INVISIBLE

c. $(20,5,-50)^w \rightarrow (5, -150, 20)^c \rightarrow k = \frac{-2}{5} = \frac{1}{2}$, $(u,v) = (2.5, -75)$

d. $(15,10,25)^w \rightarrow (10, -75, 15)^c \rightarrow k = \frac{10}{20} = \frac{2}{3}$, $(u,v) = (\frac{20}{3}, -50)$

e. $(0,0,50)^w \rightarrow (0, -50, 0)^c \rightarrow k = \frac{10}{15} = \frac{10}{0}$, $(u,v) = (NaN, NaN)$ — INVISIBLE ONES

f. $(0,0,100)^w \rightarrow (0, 0, 0)^c \rightarrow k = \frac{10}{0}$, $(u,v) = (NaN, NaN)$

References

- https://www.youtube.com/watch?v=JxwLyZ-I7vI